INCH-POUND

MIL-DTL-13139F
2 April 2010
SUPERSEDING
MIL-C-13139E (PA)
1 April 1974

DETAIL SPECIFICATION

CHARGE, DEMOLITION, SHAPED, 15 LB.–M2A4 PARTS FOR

Reactivated after 2 April 2010 and may be used for new design and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the parts for one type of demolition item designated as Charge, Demolition, Shaped, 15 lb.– M2A4.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Method for Acceptance of Product
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8861579	-	Cone
9230162	-	Charge, Demolition, Shaped, 15 lb., M2A4, Parts Arrangement
8858399	-	Well, detonator
8858398	-	Adaptor, well, detonator
8858397	-	Detonator well assembly
8858400	-	Well, detonator, alternative
8861582	-	Cap, closing
8861568	-	Closing cap assembly
8861572	-	Cone assembly
8861570	-	Container, cone
8861569	-	Body assembly
8861567	-	Sleeve, stand-off

(Copies of these drawings are available from U.S. Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny, NJ 07806-5000, or email pica.drawing.request@conus.army.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contractual document or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Conformance inspection. A sample of items shall be subjected to conformance inspection in accordance with 4.3.

3.3 Material. Materials shall comply with the requirements on drawings cited in 4.3.2.1 – 4.3.2.9.

3.3.1 Glass. The glass, from which the cone is made, shall be clear potash-soda-lead-silicate having a minimum density of 3.00 grams per cubic centimeter when tested in accordance with 4.3.3.5.

3.4 Parts and sub-assemblies. The assemblies shall comply with all requirements specified on dwgs. 8861579 and 9230162.

3.5 Leakproofness of body assembly. The body assembly, when pressurized to 2.5 psi internally with air, shall not exhibit leakage more than that indicated by a drop of a mercury manometer differential of more than 60 percent in 5 seconds when tested in accordance with 4.3.3.1.

3.6 Leakproofness of closing cap assembly. The cap assembly shall not exhibit leakage when pressurized 2 to 3 psi internally with air for 10 seconds when tested in accordance with 4.3.3.2.

3.7 Fit of closing cap assembly. The closing cap assembly shall require a force fit to be inserted into the body assembly.

3.8 Fit of stand-off sleeve. The gap between the body and the stand-off sleeve shall not exceed 0.3 inch when a static load of 20 plus or minus 1/2 pound is applied.

3.9 Annealing of glass cone. The glass cone shall be fully annealed to relieve all undue strains which might cause failure.

3.10 Heat shock test of glass cone. The glass cone shall not break or show cracks when tested for resistance to heat shock.

3.11 Workmanship. All parts and assemblies shall be clean and free of explosive on exterior, chips, cracks, crazes, porosity, dirt, grease, rust, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the cleaning agent contaminate them. Exterior surface coatings shall be continuous except for a few light scratches not exposing the base material. All required markings shall be neat and sharply defined. Well buried stones of a maximum dimension of 3/16 inch, blisters of maximum dimension of 1/8 inch and hairline blisters will be allowed. Slight discoloration of the plunger and mold surfaces will be allowed, provide they do not indicate flaws or imperfections which would adversely affect the function of the case. The shear mark shall be as near the apex of the cone as practicable.

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4. VERIFICATION

Table I. Requirements/verification cross reference matrix

Method of Verification	Classes of Verification
1- Analysis 2- Examination 3- Test	A - First article B - Conformance

Section 3 Requirements	Description	Verification Method			Verification Class		Section 4
		1	2	3	A	B	
3.1	First article	X	X	X	X		4.2
3.2	Conformance inspection	X	X	X		X	4.3
3.3	Material	X	X	X	X	X	4.3.2.1 – 4.3.2.12
3.3.1	Glass	X			X	X	4.3.3.5
3.4	Parts and sub-assemblies	X	X	X	X	X	4.3
3.5	Leakproofness of body assembly		X	X	X	X	4.3.3.1
3.6	Leakproofness of cap assembly		X	X	X	X	4.3.3.2
3.7	Fit of closing cap assembly		X		X	X	4.3.3.3
3.8	Fit of stand-off sleeve		X	X	X	X	4.3.3.4
3.9	Annealing of glass cone		X		X	X	4.3.3.6
3.10	Heat shock test of glass cone			X	X	X	4.3.3.7
3.11	Workmanship		X		X	X	4.3

4.1 Classification of Verification. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)

b. Conformance Inspection (see 4.3)

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4.2 First Article.

4.2.1 Submission. When specified, a first article sample shall be subjected to inspection in accordance with the provisions of 4.2.2 and Table II. The first article sample shall consist of ten (10) complete sets of parts and sub-assemblies.

4.2.2 Inspections to be performed. The first article samples may be subjected to all of the examinations and tests specified in Table II.

4.2.2.1 Rejection. The first article sample shall be rejected if any of the inspections/tests fail to meet specification or drawing requirements.

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Table II. First article inspection

Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Material (see Note 1)	10	3.3	Dwgs.8861579/9230162 and all referenced drawings therein 4.3.3.5
Parts and Sub-assemblies (see Note 2)	10	3.4	4.3.2
Leakproofness of body assembly and closing cap assembly	10	3.5/3.6	4.3.3.1/4.3.3.2
Fit of closing cap assembly	10	3.7	4.3.3.3
Fit of stand-off sleeve	10	3.8	4.3.3.4
Annealing of glass cone	10	3.9	4.3.3.6
Heat shock test of glass cone	10	3.10	4.3.3.7
Workmanship	10	3.11	Visual
Note 1: Certificate of analysis shall be provided for all materials.			
Note 2: 100% inspection of all drawing characteristics shall be performed.			

4.3 Conformance Inspection.

4.3.1 Inspection lot formation. Each lot of shaped charges shall contain:

a. Body assemblies from not more than one lot interfix number from one manufacturer.
b. Glass cones from not more than one batch from one manufacturer.

4.3.2 Examination. Unless otherwise specified in the Conformance Inspection, sampling plans for major and minor defects shall be in accordance with MIL-STD-1916, Verification Level IV for Major defects and Verification Level II for Minor defects.

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4.3.2.1	Well, detonator			Drawing number 8858399 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Diameter of cavity, minimum (min.)	IV	3.4	Gage
102	Depth of cavity, min.	IV	3.4	Gage
103	Outside diameter of body, maximum (max.)	IV	3.4	Gage
Minor				
201	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.2	Adapter, well, detonator			Drawing number 8858398 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None Defined			
Major				
101	Pitch diameter of thread, min.	IV	3.4	Gage
102	Minor diameter of thread, max.	IV	3.4	Gage
Minor				
201	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.3	Detonator well assembly			Drawing number 8858397 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None Defined			
Major				
101	Concentricity of inside diameter with thread	IV	3.4	Gage
Minor				
201	Assembled length of adapter, min.	II	3.4	Gage
202	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.4	Well, detonator (alternative)			Drawing number 8858400 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical				
	None defined			
Major				
101	Pitch diameter of thread, min	IV	3.4	Gage
102	Minor diameter of thread, max.	IV	3.4	Gage
103	Diameter of small cavity, min.	IV	3.4	Gage
104	Depth of cavity, min.	IV	3.4	Gage
105	Outside diameter of small end, min.	IV	3.4	Gage
106	Depth of thread cavity, min.	IV	3.4	Gage
107	Concentricity of inside diameter with thread	IV	3.4	Gage
Minor				
201	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.5	Cap, closing			Drawing number 8861582 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major	None defined			
Minor				
201	Internal depth, min.	II	3.4	Gage
202	Flange improperly formed	II	3.4	Visual
203	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.6	Closing cap assembly			Drawing number 8861568 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Concentricity of inside diameter of cap with thread	IV	3.4	Gage
102	Leakproofness of closing cap assembly	100%	3.6	Gage
Minor				
201	Color coding improper	II	3.4	Visual
202	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.7	Cone			Drawing number 8861579 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Concentricity of large outside diameter with basic diameters	IV	3.4	Gage
102	Location of large basic diameter	IV	3.4	Gage
103	Excessive wall thickness variation	IV	3.4	Gage
104	Material improper for type	IV	3.4	Visual/Manual
105	Location of small basic diameter	IV	3.4	Gage
106	Annealing of cone	100%	3.9	Gage, 4.3.3.6
107	Heat shock test of cone	IV	3.10	4.3.3.7
Minor				
201	Large outside diameter, max.	II	3.4	Gage
202	Total height, max.	II	3.4	Gage
203	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.8	Cone assembly			Drawing number 8861572 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Mismatch at ring bottoms excessive	IV	3.4	Gage
102	Cone loose	IV	3.4	Manual
Minor				
201	Protective coating missing	II	3.4	Visual
202	Evidence of poor workmanship	II	3.11	Visual
Note: Package should be opened if there is evidence of crushing.				

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4.3.2.9	Container, cone			Drawing number 8861570 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Location of upper basic diameter	IV	3.4	Gage
102	Concentricity of small inside diameter with basic diameter	IV	3.4	Gage
103	Large inside diameter	IV	3.4	Gage
104	Location of lower basic diameter	IV	3.4	Gage
Minor				
201	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.10	Body assembly, prior to attaching carrying straps			Drawing number 8861569 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Concentricity of container of cone	IV	3.4	Gage
102	Cone assembly insecure	IV	3.4	Manual
103	Reinforcing ring missing or improperly crimped	IV	3.4	Visual
104	Leakproofness of body assembly	100%	3.5	Visual
105	Fit of closing cap assembly	25	3.7	Manual
Minor				
201	Gap at base, max.	II	3.4	Gage
203	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.11	Body assembly			Drawing number 8861569 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Bare spot on interior surface	IV	3.4	Visual
Minor				
201	Color code improper	II	3.4	Visual
202	Carrying strap missing or improperly assembled	II	3.4	Visual/Manual
203	Evidence of poor workmanship	II	3.11	Visual

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4.3.2.12	Sleeve, stand-off			Drawing number 8861567 Next higher assembly 9230162
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Total length, max.	IV	3.4	Gage
102	Distance shoulder to end	IV	3.4	Gage
103	Fit of stand-off sleeve	25	3.8	Gage
Minor				
201	Color code improper	II	3.4	Visual
202	Evidence of poor workmanship	II	3.11	Visual

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4.3.3 Test Methods and Procedures.

4.3.3.1 Leakproofness of body assembly. The unit shall be assembled to the test equipment, the required internal pressure applied, and observation made for leakage for the specified period. Any assembly exhibiting more than the specified leakage shall be classed defective and removed from the lot. (Non-Destructive Test)

4.3.3.2 Leakproofness of closing cap assembly. The unit shall be assembled to the test equipment, the required internal pressure applied, and observation made for leakage for the specified period. Any assembly exhibiting leakage shall be classed defective and removed from the lot. (Non-destructive test)

4.3.3.3 Fit of closing cap assembly. Unless otherwise noted, a sample of twenty-five (25) closing cap assemblies and twenty-five (25) body assemblies minus cone assembly shall be selected from each lot for this test. The samples of closing cap assemblies and cone containers shall be segregated into combinations consisting of one unit of each. The closing cap assembly shall be assembled to its cone container using applicable equipment. Failure to require a force fit shall class a combination defective. Failure of one or more combinations in meeting the requirement shall reject both lots of components. (Non-destructive test)

4.3.3.4 Fit of stand-off sleeve. Unless otherwise noted, a sample of twenty-five (25) stand-off sleeves and twenty-five (25) body assemblies shall be selected from each lot for this test. The samples of stand-off sleeves and body assemblies shall be segregated into combinations of one unit of each. The stand-off sleeve shall be placed on a level surface, the body assembly placed thereon, and the required load applied to the top of the body assembly. The gap between the components shall be measured, the assembly picked up and given a sharp shake toward ejection of the sleeve. If the gap is excessive or the sleeve disassembles when shaken, the combination shall be classed defective. Failure of one or more combinations in meeting either requirement shall reject both lots of components. (Non-destructive test)

4.3.3.5 Density of glass cone. Ten (10) sample cones in each lot shall be checked to determine compliance with the density requirements. The density shall be obtained by weighing the sample cone, first in air and then, while freely suspended in distilled water. The density shall be calculated by means of the following formula:

$$\frac{Wa}{Wa - Ww} = Density$$

where:

Wa = weight of cone in air
Ww = weight of cone in water

If any sample cone fails to meet this requirement, this failure shall be considered cause for rejection of the lot represented. (Non-destructive test)

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4.3.3.6 Annealing of glass cone. The cones shall be examined by means of a polariscope to determine whether the glass has been relieved of all strains. Cones showing evidence of unrelieved strains shall be rejected. Any glass cone which fails to comply with the applicable requirements shall be classed defective and removed from the lot. (Non-destructive test)

4.3.3.7 Heat shock test of glass cone. The sample cone shall be conditioned at 7 degrees plus or minus 2 degrees Centigrade (C) for twenty (20) minutes minimum. The convex surface of the cones shall then be placed for two minutes, minimum, into water maintained at 95 degrees plus or minus 5 degrees Centigrade (C). Observation shall be made for compliance with the applicable requirement. (Non-destructive test)

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The 15 lb. Shaped Demolition Charge is primarily employed for burring holes in earth, metal, concrete, and paved or unpaved roads.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Requirements for Ammunition Lot Numbering, if applicable.

d. Requirement for acceptance inspection equipment (AIE) designs.

e. Certificate of Conformance for each lot or shipment of product.

f. Preservation and packaging. Preservation, packaging, packing and marking should be in accordance with Dwg. 9281170.

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6.3 Submission of inspection equipment designs for approval. Equipment designs should be submitted per the Contract Data Requirements Lists (DD Form 1423).

6.4 Certification. Unless otherwise specified by the Contracting Officer, certification should be submitted by the contractor in accordance with the contract. Distribution should include: U.S. Army ARDEC, RDAR-MEM-G, Picatinny, NJ 07806-5000.

6.5 Subject term (key word) listing.

Explosive
Composition B

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes."

Custodian:
Army-AR
Navy -OS
Air Force - 70

Preparing Activity:
Army-AR
(Project 1375-2010-005)

Review Activity
Navy – AS, NP
Air Force – 11, 99

Civil agency:
GSA-FAS

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